Big Data Technologies

Course Code: CIT 421
Course Type: Theory + Practical
Year: Fourth
Semester: VIII
Credit Hour: 3 (2TH+1 PR)
Contact Hours: 45

Course Objectives:

- Understand the Big Data Platform and its Use cases
- Provide an overview of GFS and Map Reduce principles
- Provide an overview of Apache Hadoop
- Provide HDFS Concepts and Interfacing with HDFS
- Understand Map Reduce Jobs in Hadoop
- Provide hands on Hadoop Ecosystem
- Apply analytics on Structured, Unstructured Data.
- Provide an overview of Indexing and Searching

Course Description:

Big Data is the hot new buzzword in IT circles. The proliferation of digital technologies with digital storage and recording media has created massive amounts of diverse data, which can be used for marketing and many other purposes. The concept of Big Data refers to massive and often unstructured data, on which the processing capabilities of traditional data management tools result to be inadequate.

Course Outcomes:

- Identify Big Data and its Business Implications.
- List the components of Hadoop and Hadoop Ecosystem
- Access and Process Data on Distributed File System
- Manage Job Execution in Hadoop Environment
- Develop Big Data Solutions using Hadoop Ecosystem

Course Contents:

UNIT I: Introduction To Big Data And Hadoop [8 hrs]

Types of Digital Data, Introduction to Big Data, Big Data Analytics, Role of Distributed System in Big Data, Current trends in Big Data Analytics. Structure and Unstructured Data, Introduction to Nosql databases.

UNIT II: Google File System and Map Reduce Framework [10 hrs]

GFS: Architecture of GFS, read, write, append algorithms in GFS, Fault tolerance in GFS.

Map Reduce: Anatomy of a Map Reduce Job Run, Architecture, Fault tolerance in Map Reduce, Job Scheduling, Shuffle and Sort, Task Execution, Parallel efficiency of MapReduce,Map Reduce Features.

UNIT III: Apache Hadoop And HDFS (Hadoop Distributed File System) [10 hrs]

History of Hadoop, Apache Hadoop, Analyzing Data with Unix tools, Analyzing Data with Hadoop, Hadoop Streaming, Hadoop Ecosystem.

The Design of HDFS, HDFS Concepts, Command Line Interface, Hadoop file system interfaces, Data flow, Data Ingest with Flume and Scoop and Hadoop archives, Hadoop I/O: Compression, Serialization Avro and File-Based Data structures.

Unit IV : Hadoop EcoSystem [10 hrs]

Pig : Introduction to PIG, Execution Modes of Pig, Comparison of Pig with Databases,Grunt, Pig Latin, User Defined Functions, Data Processing operators.

Hive : Hive Shell, Hive Services, Hive Metastore, Comparison with Traditional Databases, HiveQL, Tables, Querying Data and User Defined Functions.

Hbase : Basics, Concepts, Clients, Example, Hbase Versus RDBMS.

Spark : Introduction and Basic concepts

UNIT V : Searching And Indexing [7 hrs]

Full Text Indexing and Searching, Indexing with Lucene, Introduction to Apache Solr,Distributed search with Elasticsearch. Data store, list, update index and delete with elasticsearch.

Practical

1. **Installation of Apache Hadoop and familiar with HDFS**
   Setup HDFS in a single node to multi node cluster, perform basic file system operation on it using commands.
2. **Map-Reduce on Apache Hadoop**
   Write various MR programs dealing with different aspects of it.
3. **Hbase**
   Setup of Hbase in single node and distributed mode, write a program to write inHbase and query it.
4. **Elasticsearch**
   Setup elastic search in single and distributed mode, define template, write data in itand finally query it.
5. **Apache Solr**
   Setup Apache Solr in single, define template, write data in it and finally query it.

Textbook(s):

- Tom White “ Hadoop: The Definitive Guide” Third Edit on, O’reily Media.
- Seema Acharya, Subhasini Chellappan, "Big Data Analytics" Wiley.

References:

- Jay Liebowitz, “Big Data and Business Analytics” Auerbach Publications, CRC press (2013)
- Tom Plunkett, Mark Hornick, “Using R to Unlock the Value of Big Data: Big Data Analytics with Oracle R Enterprise and Oracle R Connector for Hadoop”,McGraw-Hill/Osborne Media (2013), Oracle press.
- Pete Warden, “Big Data Glossary”, O’Reily, 2011.
- Michael Mineli, Michele Chambers, Ambiga Dhiraj, "Big Data, Big Analytics: Emerging Business Intelligence and Analytic Trends for Today's Businesses", Wiley Publications, 2013.
- Paul Zikopoulos, Dirk DeRoos, Krishnan Parasuraman, Thomas Deutsch, James Giles, David Corigan, "Harness the Power of Big Data The IBM Big Data Platform ", Tata McGraw Hill Publications, 2012.
- https://research.google.com/archive/mapreduce-osdi04.pdf